

Design & Simulate a Multi-Network Infrastructure

Part 1: Network Design & IP Addressing Plan

1.1 Project Overview

This project focuses on designing and implementing a structured network for a **Food Distribution Company**. The network is designed to ensure high availability, security, and efficient data flow between different departments. The infrastructure is divided into five distinct subnets to manage traffic and enhance security policies.

1.2 Departmental Breakdown

The company’s operations are organized into the following functional areas:

- **Finance:** Handles sensitive accounting data and payroll.
- **Sales:** Manages high-volume customer orders and transactions.
- **Admin:** Supports general corporate operations and human resources.
- **IT:** Dedicated to network management and technical support.
- **Servers:** A centralized area hosting essential services like **HTTP** (Web) and **DNS**.

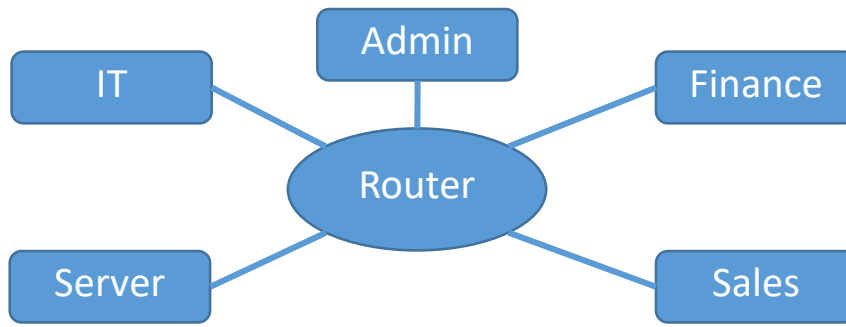
1.3 IP Addressing Table

The following table outlines the **IPv4** addressing scheme used for the project:

Department (VLAN)	Network Address	Subnet Mask	Default Gateway
Finance	192.168.10.0	255.255.255.0	192.168.10.1
Sales	192.168.20.0	255.255.255.0	192.168.20.1
Admin	192.168.30.0	255.255.255.0	192.168.30.1
IT	192.168.40.0	255.255.255.0	192.168.40.1
Servers	192.168.50.0	255.255.255.0	192.168.50.1

1.4 Network Diagram

The proposed network topology is designed using a **Star-Hierarchical** approach. In this design, a central **Cisco 2911 Router** serves as the core of the infrastructure, interconnecting three primary directions (Top, Left, and Right). Each direction is managed by a switch that supports one or more departments through logical segmentation.



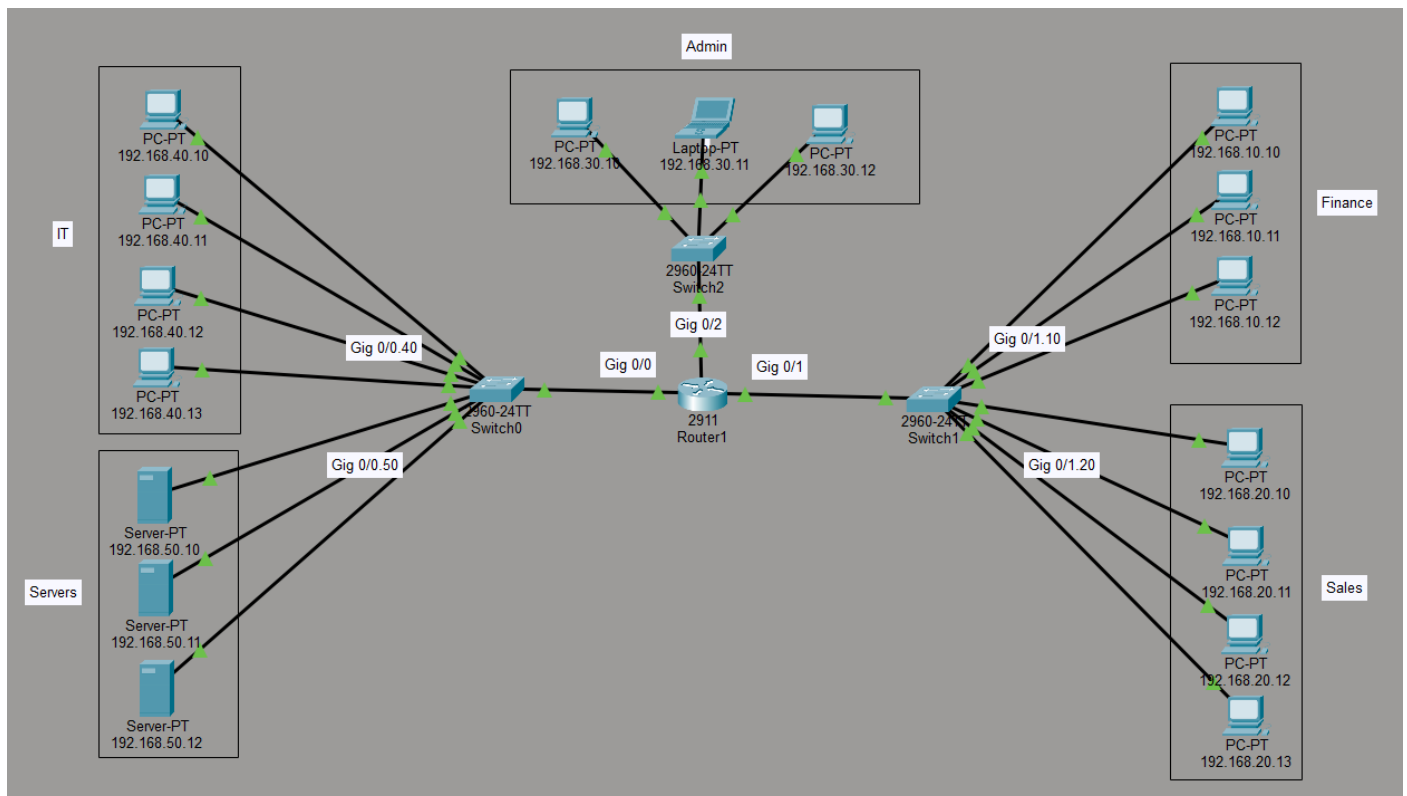
1.5 Design Justification

- **Easy to Expand (Scalability):** Using separate switches for each side makes it easy to add more computers in the future without disturbing the rest of the network.
- **Better Organization (Logical Segmentation):** We used "Sub-interfaces" on the router to keep departments like **Finance** private and safe from general traffic, while still allowing them to talk to other units when needed.
- **Smart Use of Tools (Resource Efficiency):** This design uses the router's high-speed ports effectively to make sure everyone can access the **Server Farm** quickly and without delay.

Part 2 – Network Implementation

2.1 Building the Topology

The physical network was constructed in **Cisco Packet Tracer** using a **Cisco 2911 Router** and three **Cisco 2960 Switches**. The organization is divided into five logical units: **Admin, IT, Servers, Finance, and Sales**. Each unit is connected to the central router to ensure a structured star topology.



2.2 Assigning IP Addresses

Following the predefined addressing plan, static IP addresses and Subnet Masks were assigned to all end-devices. Each PC and Server was configured with its specific **Default Gateway** (the router's sub-interface IP) to allow communication outside its own network.

2.3 Router Interface Configuration

The router was configured to handle multiple networks through a single physical port using the **Router-on-a-Stick** method.

- **Encapsulation:** 802.1Q protocol was used to distinguish between different VLANs.
- **Sub-interfaces:** Separate virtual interfaces were created (e.g., G0/1.10, G0/1.20) to act as gateways for each department.

```
enable
configure terminal
interface gig0/0
no shutdown
interface gig0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
interface gig0/0.50
encapsulation dot1Q 50
ip address 192.168.50.1 255.255.255.0
interface gig0/1
no shutdown
interface gig0/1.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
interface gig0/1.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
interface gig0/2
no shutdown
ip address 192.168.30.1 255.255.255.0
exit
exit
write memory
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/0.40
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#interface gig0/0.50
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.50, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.50, changed state to up

Router(config-subif)#encapsulation dot1Q 50
Router(config-subif)#ip address 192.168.50.1 255.255.255.0
Router(config-subif)#interface gig0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface gig0/1.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface gig0/1.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#interface gig0/2
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

2.4 Switch Configuration (Left and Right)

The switches were configured to support logical separation using **VLANs**.

- **VLAN Creation:** VLANs 10, 20, 40, and 50 were created and named according to the departments.
- **Access Ports:** Specific ports were assigned to each VLAN to keep department traffic isolated.
- **Trunk Ports:** The ports connecting the switches to the router were set to **Trunk Mode** to allow all VLAN traffic to pass through.

Left Switch	Right Switch
enable configure terminal vlan 40 name IT_Department exit vlan 50 name Server_Farm exit interface range fa0/1-10 switchport mode access switchport access vlan 40 interface range fa0/11-20 switchport mode access switchport access vlan 50 interface gig0/1 switchport mode trunk exit exit write memory	enable configure terminal vlan 10 name Finance exit vlan 20 name Sales exit interface range fa0/1-10 switchport mode access switchport access vlan 10 interface range fa0/11-20 switchport mode access switchport access vlan 20 interface gig0/1 switchport mode trunk exit exit write memory

```
Switch0
Physical Config CLI Attributes
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name IT_Department
Switch(config-vlan)#exit
Switch(config)#vlan 50
Switch(config-vlan)#name Server_Farm
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#interface range fa0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 50
Switch(config-if-range)#interface gig0/1
Switch(config-if-range)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Switch#
```

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Top

```
Switch1>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Finance
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface range fa0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#interface gig0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Switch#
```

2.5 Connectivity Testing (Ping Results)

To verify the implementation, extensive connectivity tests were performed using the **ICMP (Ping)** command. Successful pings from the Finance PC (192.168.10.10) to other networks confirmed that the router is correctly routing traffic between all five departments.

- Ping to IT: **Successful**
- Ping to Sales: **Successful**
- Ping to Admin: **Successful**
- Ping to Servers: **Successful**

```
192.168.10.10
Physical Config Desktop Programming Attributes
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Part 3 – Protocol & Communication Analysis

3.1 Role of the Router

In our Food Distribution Company network, the **Cisco 2911 Router** acts as the central brain. Its main role is to connect different local networks (VLANs) that cannot talk to each other directly. It manages the flow of data and ensures that a packet from the **Finance** department reaches the correct **Server** in the server farm.

3.2 Purpose of the Default Gateway

The **Default Gateway** is the "exit door" for every computer in our network. Each department has its own gateway IP (e.g., 192.168.10.1 for Finance). When a PC wants to send data to a different department, it sends the data to this gateway. Without it, devices could only communicate with others in the same room (same VLAN).

3.3 How Routing Works (TCP vs. UDP)

Routing is the process of finding the best path for data.

- **TCP (Transmission Control Protocol):** It is reliable and ensures data arrives correctly. For example, when an **IT staff** member accesses a **database on the Server**, TCP is used. It creates a "handshake" to make sure the server is ready before sending the data.
- **UDP (User Datagram Protocol):** It is faster but doesn't check if data arrived. It is used for things like live video streaming where speed is more important than 100% accuracy.
- **In our design:** We primarily use **TCP** for department-to-server communication to ensure no data is lost during financial or administrative tasks.

3.4 Application Protocols Examples

In our company scenario, we use several protocols to handle daily tasks:

- **HTTP / HTTPS:** Used by employees to browse the company's internal website (the web portal for food orders).
- **DNS (Domain Name System):** Translates names like `www.food-dist.com` into the Server's IP address, so users don't have to memorize numbers.
- **DHCP (Dynamic Host Configuration Protocol):** Automatically gives IP addresses to new PCs added to the Sales or Finance departments.
- **FTP (File Transfer Protocol):** Used by the **IT department** to upload large firmware updates or backup files to the Main Server.
- **Email (SMTP/POP3):** Allows the **Admin** department to send official notices and invoices to other units.

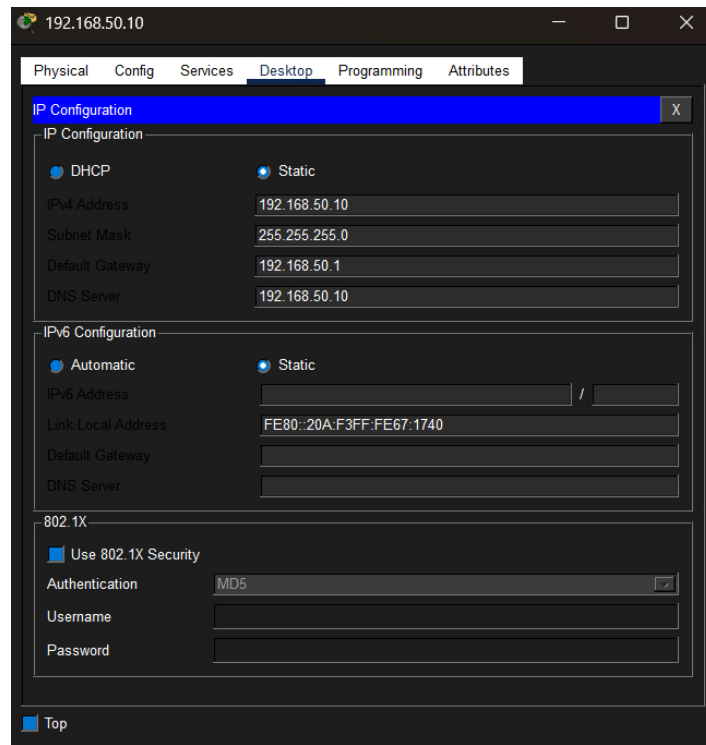
3.5 Server Services Implementation

To demonstrate the practical use of the network, we configured the central Server to provide web and naming services. The following steps were performed:

Step 1: Server IP Configuration (Prerequisite)

Before enabling services, the Server was configured with a static IP address to ensure it remains reachable by all departments.

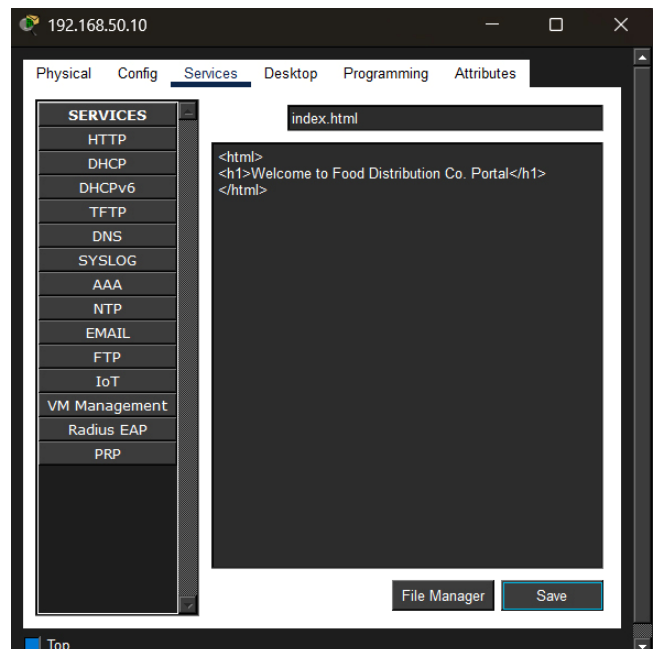
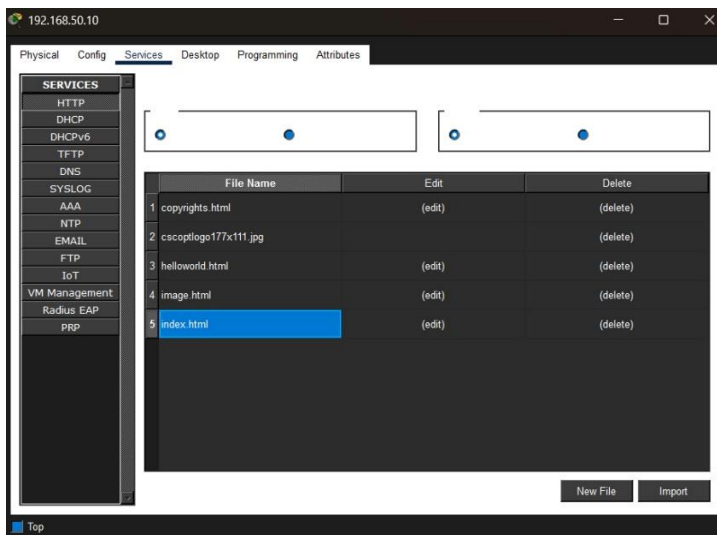
- **IP Address:** 192.168.50.10
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.50.1 (Router's sub-interface for the Server Farm)
- **DNS Server:** 192.168.50.10 (The server points to itself for DNS resolution)



Step 2: Enabling HTTP Service (Company Website)

We activated the Web Server service to host the company's internal portal.

- Navigated to **Services > HTTP** and turned **On** both HTTP and HTTPS options.
- Edited the index.html file to display a custom welcome message: `<h1>Welcome to Food Distribution Co. Portal</h1>`.

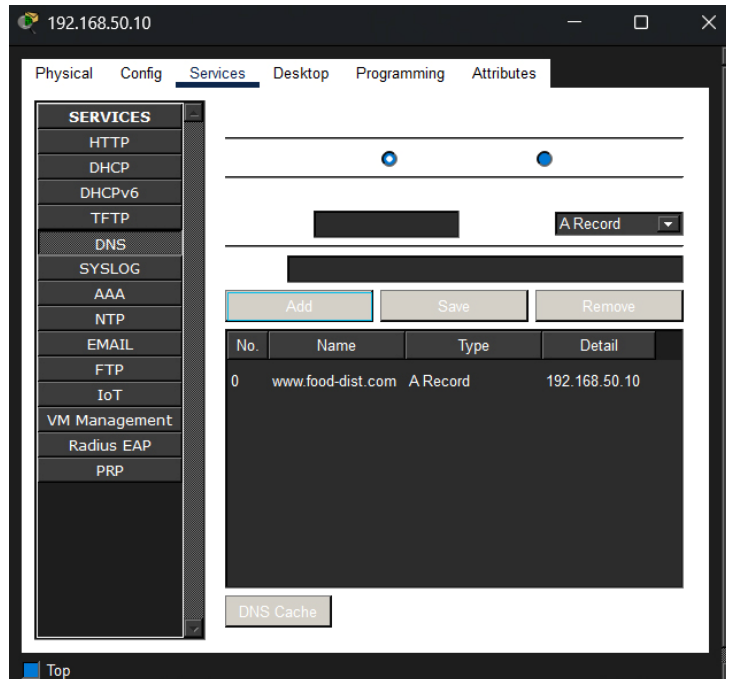
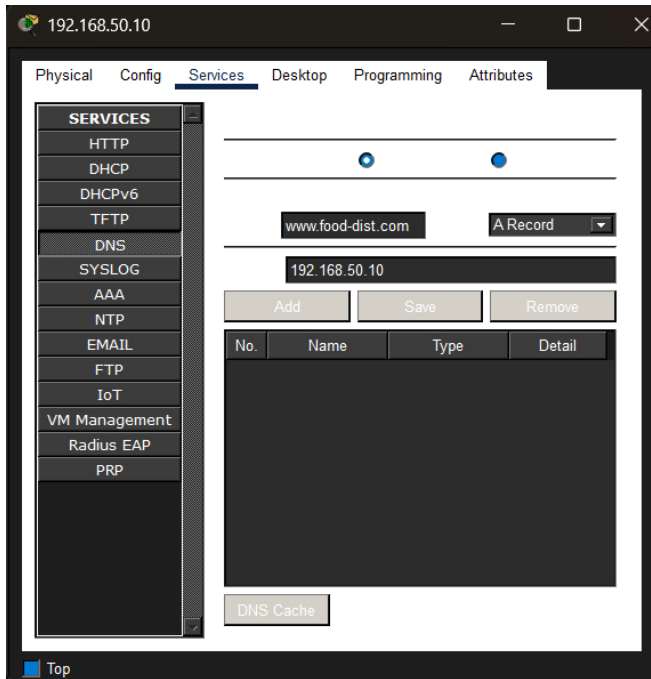


Step 3: Enabling DNS Service (Domain Name Setup)

To avoid using complex IP addresses, the DNS service was configured to map a domain name to the Server's IP.

- Navigated to **Services > DNS** and turned the service **On**.
- Added a new **A Record**:

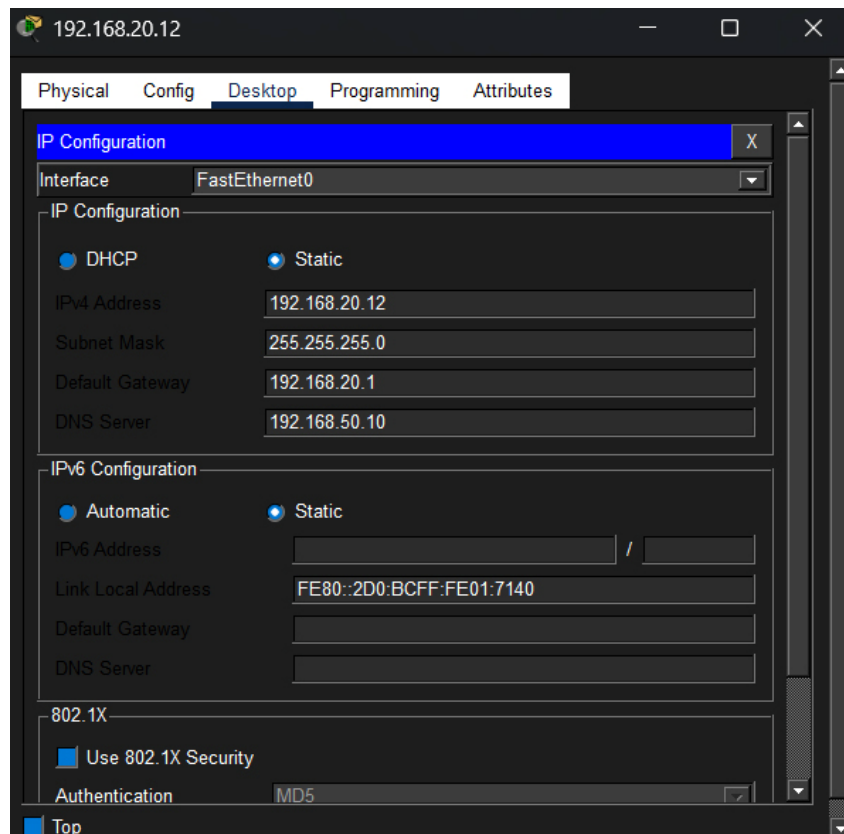
- **Name:** www.food-dist.com
- **Address:** 192.168.50.10



Step 4: Client DNS Configuration

For the PCs in other departments (like **Sales** and **Finance**) to recognize the domain name, their IP settings were updated.

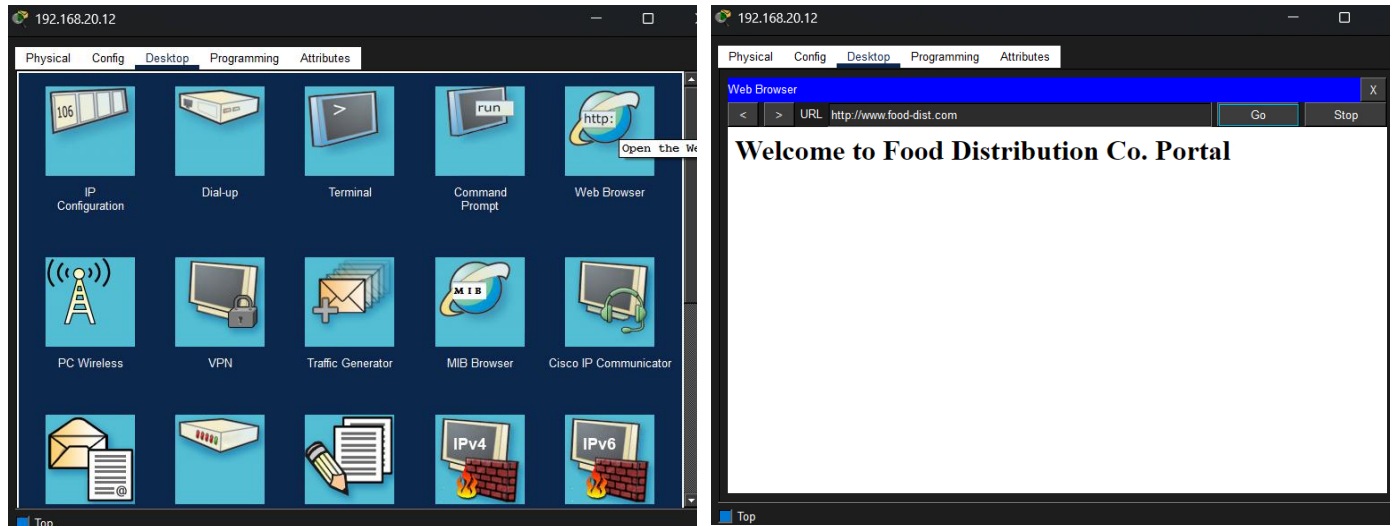
- Entered the **IP Configuration** of each PC.
- Set the **DNS Server** field to 192.168.50.10.



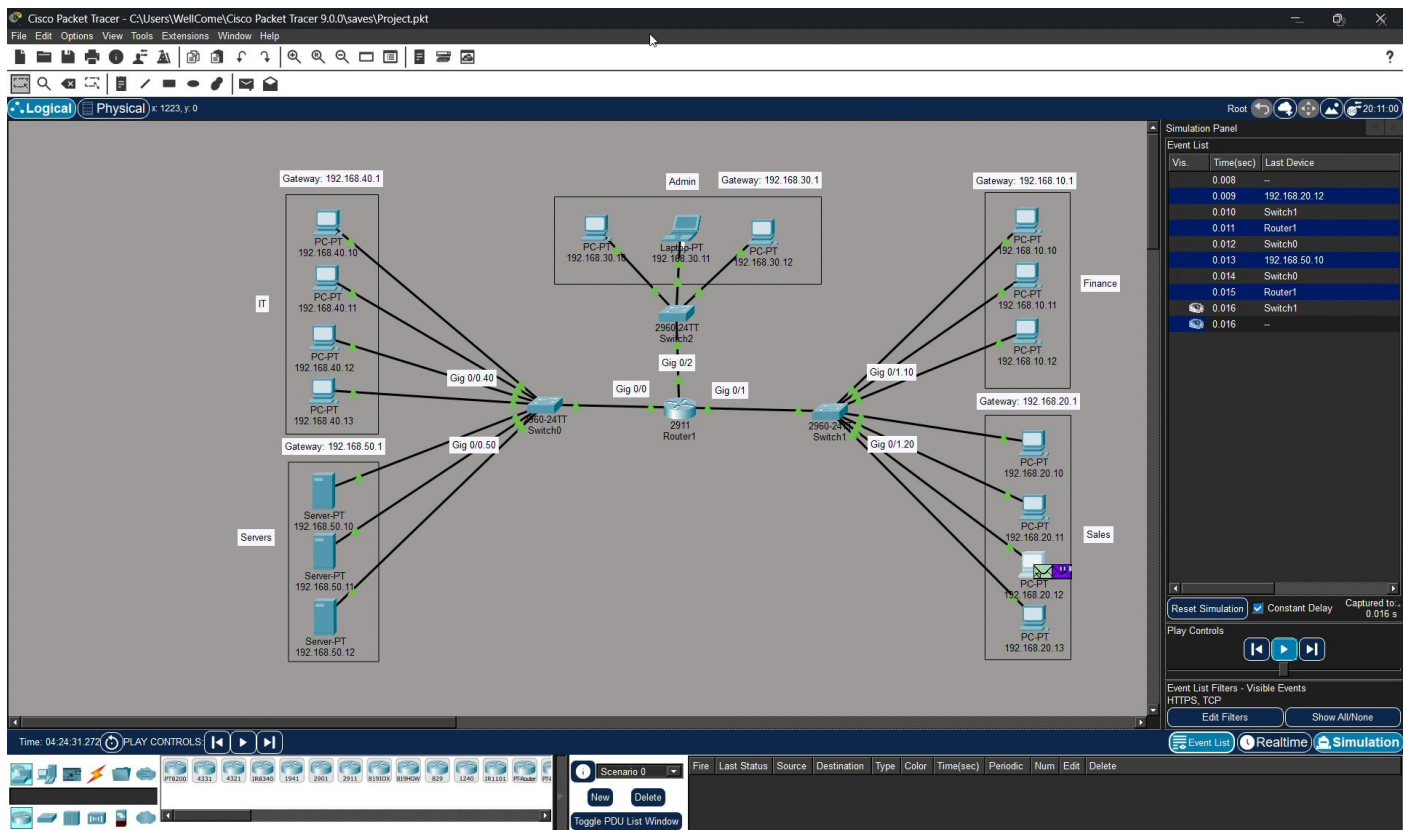
Step 5: Final Verification (The Moment of Truth)

To confirm the entire infrastructure (Routing, HTTP, and DNS) is working correctly:

1. Opened the **Web Browser** on a PC in the **Sales** department.
2. Entered `www.food-dist.com` in the address bar.
3. The custom company homepage loaded successfully, proving that inter-VLAN routing and application services are fully operational.



To further analyze the network communication, Simulation Mode was used to track TCP traffic. This confirmed the reliable delivery of data and verified that the Router correctly performs Inter-VLAN routing for every step of the TCP three-way handshake.

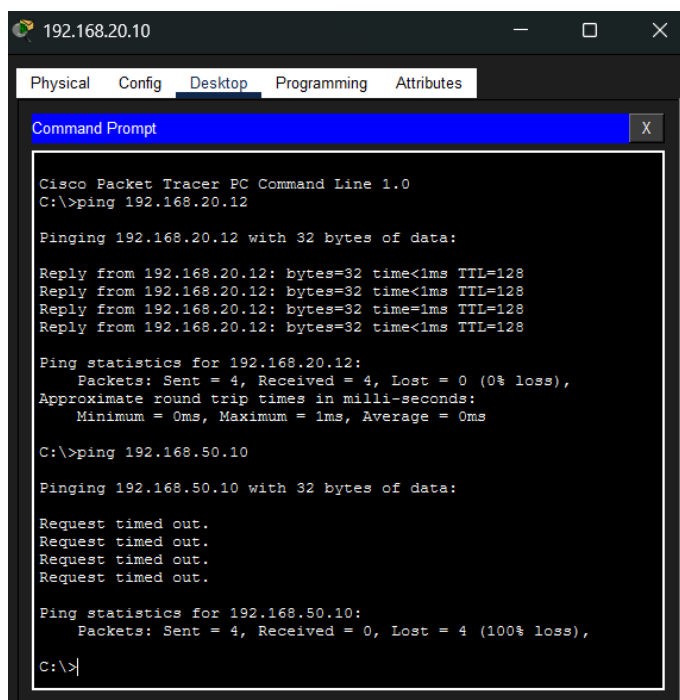


Part 4 – Troubleshooting Section

In any network environment, issues can arise due to configuration errors. This section explores two common problems encountered during the setup and their solutions.

Problem 1: Incorrect Default Gateway

- **Description:** A PC in the **Sales** department was unable to access the **Server Farm**, although it could ping other devices in the same room.
- **Investigation:** Upon checking the IP Configuration, it was discovered that the **Default Gateway** was set incorrectly to 192.168.20.5 instead of 192.168.20.1.
- **Solution:** Corrected the Gateway IP on the Sales PC. This restored the connection to the Router and the rest of the network.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128

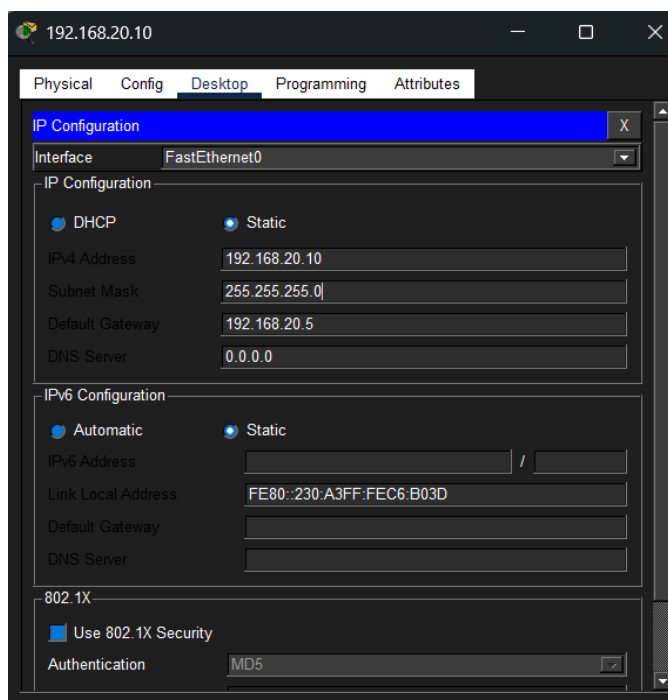
Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```



IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.5

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::230:A3FF:FEC6:B03D

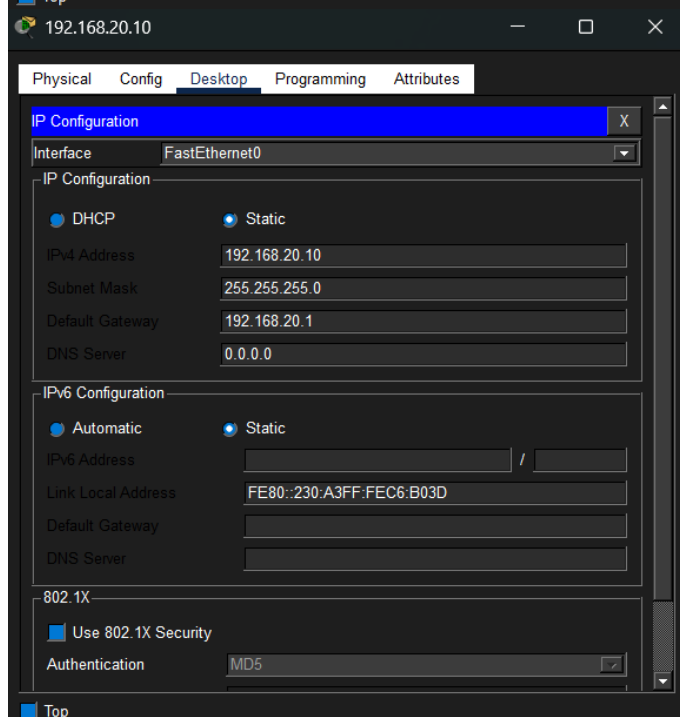
Default Gateway:

DNS Server:

802.1X

☒ Use 802.1X Security

Authentication: MD5



IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::230:A3FF:FEC6:B03D

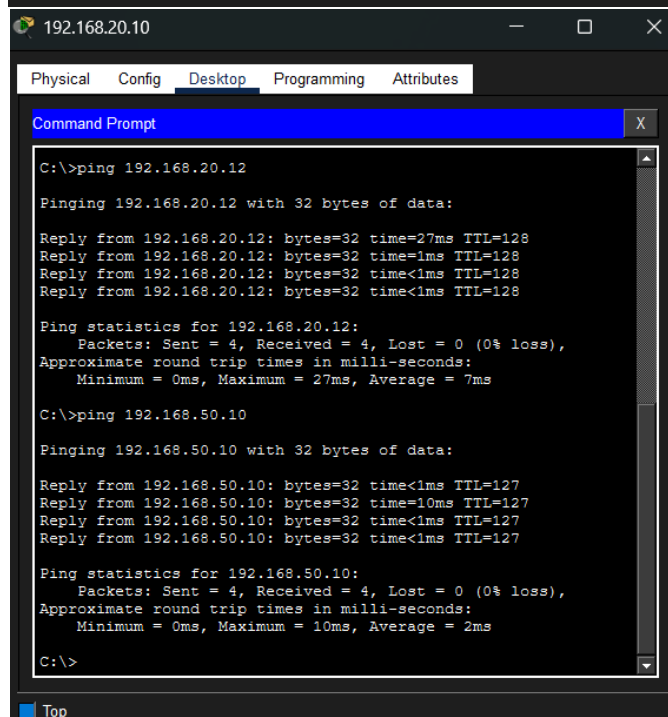
Default Gateway:

DNS Server:

802.1X

☒ Use 802.1X Security

Authentication: MD5



```
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Reply from 192.168.20.12: bytes=32 time=27ms TTL=128
Reply from 192.168.20.12: bytes=32 time=1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 27ms, Average = 7ms

C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time=10ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Problem 2: Disabled Interface (Shutdown State)

- **Description:** The entire **IT and Server** wing lost connectivity to the rest of the company.
- **Investigation:** we noticed that interface GigabitEthernet 0/0 was in an "Administratively Down" state.
- **Solution:** Entered global configuration mode and used the no shutdown command on the interface. The link status changed to "Up," and connectivity was restored.

